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Energy system modelling and GIS to build an Integrated Climate Protection Concept for Gauteng Province, South Africa

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HIGHLIGHTS

- Two models to assess the transport sector have been developed.
- The methodology covers the energy system and locational information.
- Application to Gauteng to provide input for a Climate Protection Concept.
- Energy efficient polices will help to significantly reduce transport emissions.
- Local renewable resources and efficient powertrains should be part of this policy.

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ABSTRACT

South Africa and specifically its economically dominant province of Gauteng aim to reduce their influence on climate change. Especially the transport sector is seen as one of the key drivers of future greenhouse gas (GHG) emissions. This paper describes the methodology used to combine the application of two models in order to provide a basis for informed policy recommendation for GHG mitigation.

The TEMT model provides real world emission factors adapted to local conditions in Gauteng for numerous vehicle technology concepts. Those data feed into the TIMES-GEECO energy system model which identifies future technology use for different alternative scenarios. Finally, the scenario results are illustrated spatially using a GIS programme.

The results of the scenario analysis show that under implemented policies GHG emissions in Gauteng are likely to increase substantially. Pollutant emissions are currently high as a result of a comparably old vehicle fleet. The spatial display of these results shows where the traffic network is concentrated and the location of so-called emission hot-spots. Energy efficient policies for the transport sector of Gauteng can achieve a significant reduction of emissions and energy consumption. Alternative powertrains and the use of locally produced biofuels can play a significant role in such policies.

1. Introduction

Gauteng province is the economic powerhouse of South Africa housing a fifth of the population and generating a third of the national GDP (UN-Habitat, 2010). Gauteng's population is expected to grow at a relatively high rate, even faster than other parts of the country (Wehnert et al., 2010). The socio-economic growth is the driving force behind the increasing energy demand. Given its economic dominance in South Africa, it is not surprising that Gauteng is responsible for about one third of the total final energy

Abbreviations: BRT, Bus Rapid Transit; BTL, Biomass-to-liquid; CCS, Carbon capture and storage; CNG, Compressed natural gas; CTL, Coal-to-liquid; GDP, Gross domestic product; GHG, Greenhouse gas; GIES, Gauteng Integrated Energy Strategy; GIS, Geographic Information System; GTL, Gas-to-liquid; HBEFA, The Handbook of Emission Factors for Road Transport (Handbuch für Emissionsfaktoren des Strassenverkehrs) for Germany, Switzerland and Austria; HC, Hydrocarbons; IGCC, Integrated gasification combined-cycle; IPO, Implemented Policies scenario; LCV, Light commercial vehicle; LGL, Low carbon province – least-cost scenario under GIES mitigation targets; LGS, Low carbon province – solar province scenario under GIES mitigation targets; LPG, Liquefied petroleum gas; LRS, Low carbon province – GHG emissions as "required by science"-scenario in the LTMS; LTMS, Long-term Mitigation Strategies; MATSim, Multi-Agent Transport Simulation; NATIS, National Traffic Information System of South Africa; NHTS, National Household Travel Survey; NMHC, Non-methane hydrocarbons; PM, Particulate matter; RES, Reference Energy System; REIPPP, Renewable Energy Independent Power Producer Procurement; SWH, Solar Water Heater; TEMT, Transport Emissions Modelling Tool; TIMES, The Integrated Markal-EFOM System; TIMES-GEECO, TIMES-Gauteng Energy and Emissions Cost Optimisation; TTW, Tank-to-wheel; VDF, Volume-delay function

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consumption of the country (Tomaschek et al., 2012a).

South Africa is a country rich in coal but has almost no indigenous oil reserves. Due to the fact that all oil refineries and most of the electricity-generating plants are located outside the province,[1] most of the energy is imported and a significant amount of GHG emissions which are attributable to provincial economic activity are generated elsewhere in the country. The total GHG emissions (in 2009) corresponding to the energy consumed in Gauteng was about 126 Mt CO_2eq, taking into account the emissions of the whole production chain of the energy used in the province where thereof only about 40 Mt CO_2eq GHG emissions are produced within Gauteng (Tomaschek et al., 2012a).

Gauteng's provincial government sees its chance to make the province a forerunner for climate protection in South Africa and looks for clear investment priorities. The Province has developed the Gauteng Integrated Energy Strategy (GIES) to propose possible measures to mitigate GHG emissions (DLGH, 2010) in response to the National Long-term Mitigation Strategies (LTMS) (Winkler, 2007). However, especially in the transport sector, there is only limited information on the performance of possible GHG mitigation options available. For example, basic statistics (e.g. fuel consumption figures) are not easily accessible (Goyns, 2008) and priorities for implementing the different measures in all sectors have not been set by policymakers.

To characterise the provincial transport sector and in order to improve policymaking to reduce energy consumption and GHG emissions in the province while focusing on the demands of society and budget constraints, we have developed two transport tools, namely, TEMT[2] and TIMES-GEECO[3]. This paper describes the methodology applied using the combination the two models assessing how the transport sector can be assessed within an energy system taking into consideration locational information (GIS) in order to provide input towards an Integrated Climate Protection Concept for Gauteng, South Africa.

2. Materials and methods

Transport emissions in cities and rural areas in view of increasing road traffic density of private transport and goods transport has been the focus of various studies conducted in the last 20 years. These sought to reflect the real burden of traffic emissions in urban areas with appropriate models in the field of transport emissions.

In the European countries different emission models for urban areas were developed. Most of these emission models are generally based on the emission factors from the Handbook of Emission Factors for Road Transport (HBEFA) (Borge et al., 2014). Colberg et al. (2005) investigated the differences between calculated vehicle emissions and measured emissions in the area of the Gubrist road tunnel in Switzerland. The study concluded that modelling and calculating emissions produce comparable results. Franco et al. (2013), who gave a comprehensive review of different methods to calculate the emissions of road vehicle in urban areas, however, pointed out that it is critical to adapt the emission factors to the specific traffic situation.

MATSim (Multi-Agent Transport Simulation) software was developed in order to better reflect the real traffic situation on road transport by simulating real traffic flows in the street networks. Examples of the simulation method and of applications of this software to different traffic situations can inter alia be found in Grether and Nagel (2013) and Schroeder et al. (2012). Emission models can also be used in combination with Geographical Information Systems (GIS) to generate clean air plans for cities. Different GIS systems were used worldwide in the areas of cities, regions and landscapes to visualize results of transport flow data or transport emission data (Asdrubali et al., 2013; Kurka et al., 2012; Rodrigues et al., 2015).

Energy system models are largely applied to identify suitable technologies for balancing energy supply and demand in a cost-optimal way. Nakata et al. (2011) give an excellent overview of the energy system modelling approach as well as general technology options for GHG mitigation within an energy system including the transport sector. The International Energy Agency summarized more than 350 publications between 2008 and 2010 using the TIMES-MARKAL model generator (IEA, 2011). However, the focus of those publications clearly lies on the European or US American context and only about 6% of the publications investigated a regional or local context. Furthermore, little focus has been given on transport orientated analyses, which clearly underlines the need for research in that area.

Recent publications about energy system models specifically covering the transport sector focussed on aspects relevant for industrialised countries, such as hydrogen supply and use (see e.g. Hedenus et al. (2010) and Contreras et al. (2009)), alternative fuels for personal transport (see e.g. Gül et al., 2009) and /or electric mobility (see e.g. Densing et al. (2012)). However, some transport oriented approaches have demonstrated the prospects of energy system modelling for developing countries like China (Liu et al., 2013) or Nepal (Shakya and Shrestha, 2011).

We developed the TEMT model to create a transport emissions inventory for the province. This software model incorporates emission factors for existing vehicles as well anticipates potential alternative technologies and fuels based on available local data and the European HBEFA. Vehicle technology was defined in TEMT in terms of the size of the engine, emission class, the specific energy consumption and emission factors for different traffic situations; these were aggregated and incorporated into TIMES-GEECO. The latter is an energy system model application for South Africa focussing on Gauteng, which can identify least-cost measures to achieve the climate and energy efficiency goals of the region by integrating proposed energy policies and technologies within a defined technical and socio-economic framework. This integration includes the transport sector but also the other components of the energy system. TIMES-GEECO derives the vehicle kilometres corresponding to the technology in place and fuel consumed by each transport mode. Using this model, we conducted a scenario analysis to identify robust measures to reach provincial targets at minimum cost. A robust measure, in our case, means that it occurs in most of the future scenarios independently of the scenario assumption (e.g. future crude oil price, socio-economic growth). A scenario framework was developed for the period from 2007 to 2040 to evaluate the implications of different prospective techno-economic pathways.

The results of the TIMES-GEECO model are based on scenario calculations and display different possible energy futures for the province. These are used as input to the TEMT model. In TEMT the total vehicle activity is assigned to individual vehicles for different traffic situations to visualise the travel demand and emissions spatially using a GIS (geographic information system) programme and the simulation model MATSim. The spatial display of the scenario results allow us to show where the traffic activity is concentrated in relation to current and possible future transport needs, and the consequences for energy demand and GHG emissions.

The concept of combining GIS software with decision support tools is not new. For example, Van den Broek et al. (2010)

[1] Gauteng contributes only about 3% of the South African generation capacity (Platts, 2011).

[2] Transport Emission Modelling Tool.

[3] TIMES-Gauteng Energy and Emissions Cost Optimisation.

combined ArcGIS and a MARKAL application for the Netherlands to optimize a possible infrastructure for CO_2 transport. A combination of GIS and a mixed-integer optimization approach to identify optimal utilization of sewage sludge was demonstrated by Vaskan et al. (2013) for the Catalonian area in Spain and Strachan et al. (2009) presented a methodology to assess proposed hydrogen infrastructure in the UK.

However, this is the first time that energy system modelling and GIS have been combined as a planning tool for the transport sector and for Gauteng, which allows for a better understanding of the transport sector and contributes to a more informed policy-making process.

2.1. The TEMT tool for building an emissions inventory for Gauteng and as interface to GIS software

The TEMT emission model was created by TÜV Rhineland, Cologne to generate real-world emission factors for Gauteng and to visualise transport emissions spatially. TEMT operates using Visual Basic code. It is a combination of a transport emission tool based on European research data on vehicle emissions and a Geographic Information System (ArcGIS) to visualise vehicle travel based on different vehicle types and their specific emissions directly on a road map.

The TEMT model is founded on several databases that provide primary input data:

- The *Handbuch Emissionsfaktoren* (Handbook of Emission Factors for Road Transport) for Germany, Switzerland and Austria (HBEFA) (HBEFA 2.1 (2004); HBEFA 3.1 (2010)).
- The National Traffic Information System of South Africa (NATIS) vehicle registration database for calibration of the base years of 2007–2010 (DoT, 2012).
- The MATSim (Multi-Agent Transport Simulation) software for spatial distribution of vehicle travel.
- The spatial road network data for Gauteng for application in GIS software (ArcGIS 9.2).
- TIMES-GEECO energy system model for vehicle fleet composition for future years.

We extracted the emission factors from HBEFA for common traffic situations (specifically, road type and vehicle speed) for different vehicle types with their respective emission performance for diesel and petrol engines. The HBEFA database represents the emission factors, derived from the results on test-beds, for different vehicle classes – such as passenger cars, light commercial vehicles (LCV), trucks, buses, minibuses and motorcycles – in different traffic situations and on different road-type classes at different speeds. The vehicle emission data were extended by us to alternative technologies (such as LPG engines, CNG engines, hybrid vehicles) and possible future technologies in the transport sector. As a result, we extracted emission factors for the pollutants CH_4, CO, CO_2, HC (hydrocarbons), NO_x, N_2O, NH_3, NMHC (non-methane hydrocarbons), PM (particulate matter), and SO_2 differentiated by vehicle class, engine size, weight-fuel, road types and speed.

The emissions factors and vehicle fleet were adjusted to South African conditions. We separated the baseline vehicle fleet for Gauteng for the year 2007–2009 by engine size and weight based on the NATIS vehicle registration database. The NATIS database (DoT, 2012) contains information on the type of vehicles registered, their fuel type (petrol/diesel), engine size and year of registration. The existing vehicle types from the HBEFA and the new developed alternative vehicle types were dedicated to the vehicle types from the registration database NATIS, by aligning similar fuel type, cylinder capacity and emission class (Fig. 1).

MATSim software was used to calculate the spatial distribution of vehicle travel. MATSim allows us to allocate the travel demand to vehicle volumes in peak hours by road type using volume-delay functions (VDF). The volume-delay function is an internal algorithm of the software. The volume-delay function gives travel time t as a function of link capacity fraction (Eq. (1)):

$$t(v) = t_0\left(1 + \left(\frac{v}{c}\right)^{\alpha}\right) \tag{1}$$

with t_0 = free flow travel time across the link, v = traffic volume in link, c = capacity and

α = rate of change in travel time.

MATSim data were created by the application of four basic phases in the traditional travel demand forecasting process on the basis of the South African NHTS (National Household Travel Survey) data for 2003 (DoT, 2003) with updates to the year 2009. Raw MATSim output data representing the road network and the data assigned to travel demand were transformed from tabular format into a shapefile format using Visual Basic Script in order to effect the transfer of the data to TEMT.

MATSim generates the fleet density for the different road types and the average speed of the vehicles for the roads, and thus allows the spatial distribution of the TIMES-GEECO results in GIS. To do so, we allocated the vehicle kilometres travelled by mode and technology to each road link and assigned them to a specific speed category (for example, 60 km/h). After that, we assigned the corresponding emission factors, from HBEFA and relevant to specific speed categories, for all vehicle technologies used in the scenario to each road link. Finally, the emissions for each road link were totalled and visualised using ArcGIS. In the first step the TEMT model generated the total emissions for the different layers, i.e. road types of the different speed sections and vehicle types to a typical vehicle for the chosen scenario (Eq. (2)).

$$EL_k = \sum_1^i F_i \cdot EF_i \cdot L_k,\ i=[1, \ldots, n],\ k=[1, \ldots, m] \tag{2}$$

with F_i=Vehicle quantity by of type of technology (i), EF_i = Emission factor for technology (i) (g/km), L = Link length (km) and EL = Emission per link [g].

After these calculations the visualization with ArcGIS was prepared in a grid raster on a 4 km^2 scale (Eq. (3)). In the ArcGIS model the information on the road net and the information on the emission grid can be overlapped. In this way we were able to visualise the emissions for regions, cities or other areas.

$$EG = \sum_1^k EL_k,\ k=[1, \ldots, m] \tag{3}$$

with EG = Emission in Grid (4 km^2), EL_k = Emission per link [g] and m = Quantity of links in grid.

Additionally, we created a user-friendly interface with ArcGIS 9.2 for TEMT. It allows the direct import of the modelling results to GIS and the visualization of the emissions from road transport with a predefined legend and classification scheme.

2.2. TIMES-GEECO, an energy system model for Gauteng

The TIMES-GEECO model which encompasses the entire energy system of Gauteng and the energy supply structure of South Africa, has been developed at the IER, University of Stuttgart. Detailed descriptions of the model structure is inter alia available at Haasz et al. (2013), Tomaschek et al. (2012b) and Dobbins et al. (2009). TIMES-GEECO is an application of the TIMES model generator which is a technology based partial equilibrium model that assumes perfect foresight and perfect competition (Loulou et al., 2005). This means that each technology has to be explicitly defined in terms of its technical, economic and environmental parameters (such as investment costs, conversion efficiency or

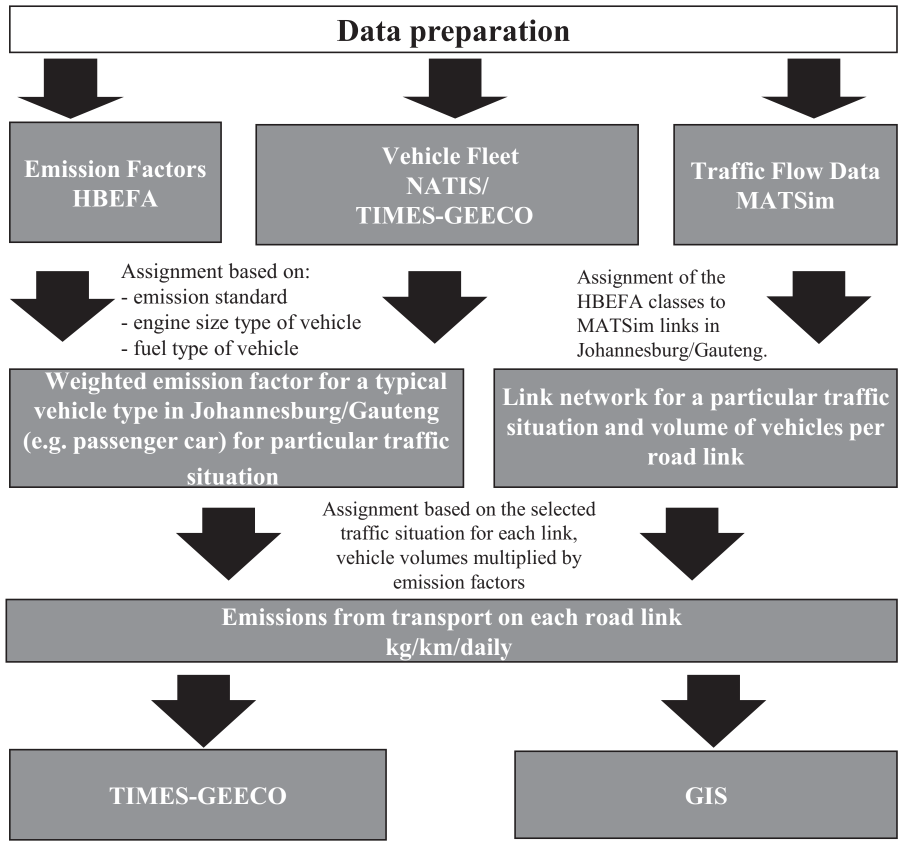

Fig. 1. Modelling approach used in TEMT to illustrate the flow of information within the model and the spatial visualisation of data.

emission factors). All relevant energy carriers at every stage of the conversion process (from primary energy provision, to secondary energy forms like electricity to end-use or energy service demands) are represented in the model and a supply and demand equilibrium exists (Loulou et al., 2005).

The standard TIMES model uses linear programming and interior point solvers for the optimization problem which can be generally described as shown in Eq. (4).

$$Obj = \sum_{k=1}^{n} c_k \bullet x_k + c \overset{!}{=} min,\ x_k \geq 0 (k = 1, \ldots, n) \tag{4}$$

where x_k are the decision variables such as activity levels, energy flows and investment decisions and c_k and c are constants.

In TIMES the objective is to minimise the total system costs (see Eq. (5)) which include capital costs $INV_{p,t}$ for investing into or dismantling a process p in time period t; fixed operating and maintenance costs and other annual costs ($FOM_{p,t}$); variable costs which are dependent on the activity (production level) of a process expressed by $ACT_{p,t}$ as well as those dependent on flow costs of commodities $FLO_{p,t}$ (which might occur e.g. due to CO_2 transport) and import expenditures and export revenues ($IMPEXP_{p,t}$). $SALV_{p,t}$ represents the salvage value of processes at the end of the planning horizon T. All costs are discounted to the beginning of the base year z using discount factor $DF(t, z)$.

$$Obj(z) = \sum_{t=1}^{T} DF(t, z) \bullet \left[\sum_{p} \left(INV_{p,t} + FOM_{p,t} + ACT_{p,t} + FLO_{p,t} + IMPEXP_{p,t} - SALV_{p,t} \right) \right] \overset{!}{=} min \tag{5}$$

Different constraints which are part of the scenario definition affect the solution of the optimization problem. Particularly significant to for our analysis is the *activity* of transport vehicle processes p within a scenario, which expresses the use of a certain technology for serving the defined demand in mobility which is then exchanged between TIMES-GEECO and TEMT.

To include all emissions and costs attributable to the provincial energy consumption in TIMES-GEECO, the entire supply sector of South Africa was fully modelled (Fig. 2). Current power plants for electricity generation were taken into account as well as residual refinery capacities – including the liquefaction of coal (CTL) and gas (GTL). Incorporating the entire energy supply chain in the modelling is deemed to be necessary as most of the life-cycle GHG emissions resulting from the manufacture of synthetic fuels in the South African fuel mix can be attributed to the provision of the energy carrier (Tomaschek et al., 2012a).

On the fuel supply side, various energy crops for biochemical biofuel production have been identified and their respective contributions quantified – these crops include sugar cane, sugar beet, rape seed, sunflower seed and soybean. Furthermore, biodiesel from waste cooking oil and ethanol from lignocellulosic biomass have been taken into account. For thermo-chemical production of liquid fuels the gasification of biomass and subsequent Fischer–Tropsch (FT) synthesis is also included (BTL). Natural gas as well as gasified and upgraded coal or solid biomass can be used to supply compressed methane gas. The model also includes hydrogen supply and usage as well as the use of organic waste to provide biogas. Additionally, we incorporated the possibility of using carbon capture and storage (CCS) for certain technologies in the model, i.e. in the electricity generation sector for coal-based Integrated gasification combined-cycle (IGCC) power plants, and in fuel liquefaction for both new CTL and GTL facilities.

Besides the primary energy supply and energy conversion

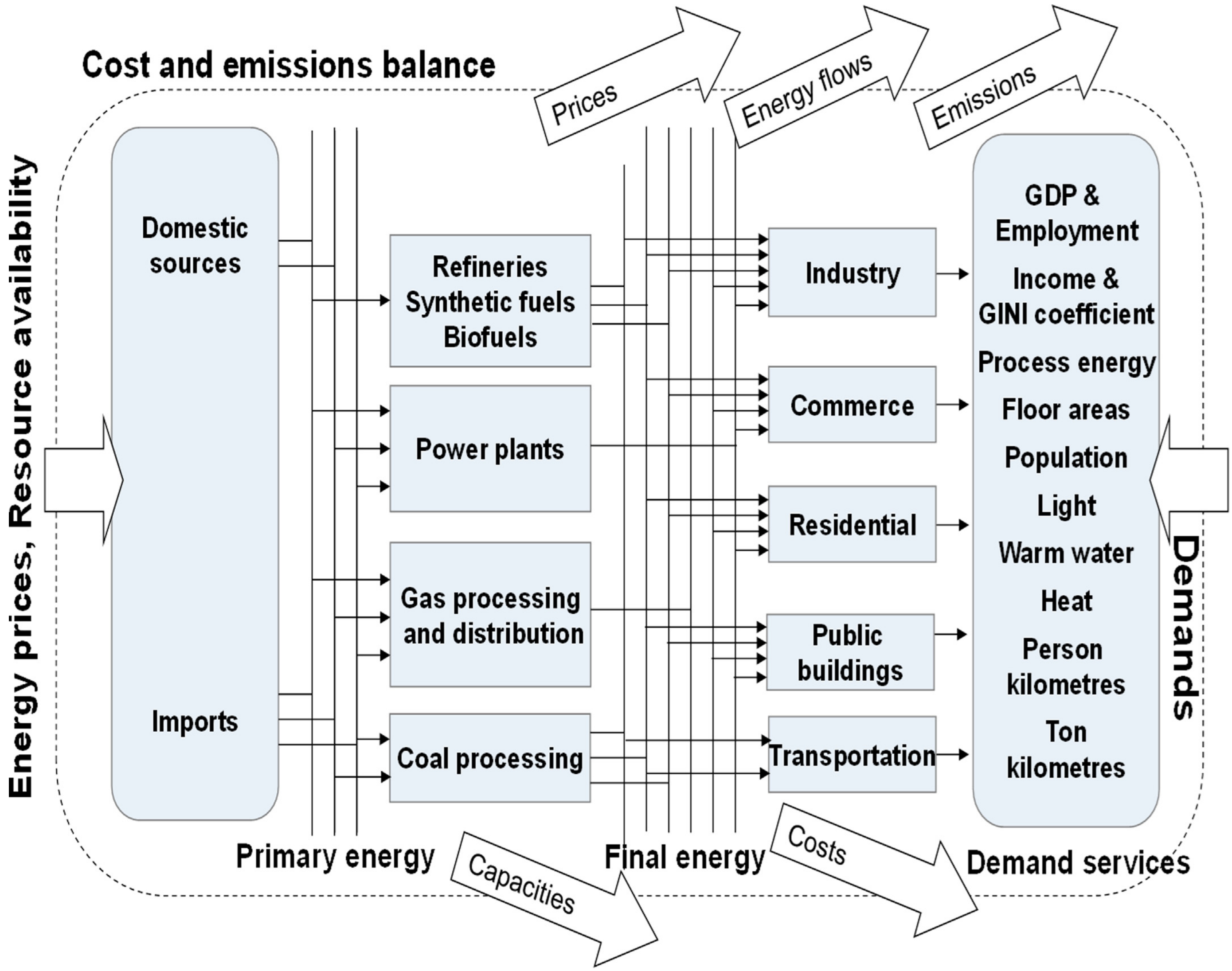

Fig. 2. Reference Energy System of the TIMES-GEECO model. The model integrates all demand sectors of the metropolitan region of Gauteng and energy supply sectors of the South African energy system. The models accounts for greenhouse gas emissions, air pollutants, energy flows, capacities and costs for all analysed sectors.

sectors, the TIMES-GEECO model includes all components of the Gauteng energy system, that is, residential, public buildings, commercial, industrial and transport (Fig. 2). The transport sector covers road, rail and air transport for individual and public passengers, as well as freight, where a wide variety of possible technology options have been implemented including conventional combustion engines, hybrid electric vehicles (mild, full, plug-in and fuel–cell), battery electric vehicles (BEV), gas engines (LPG, CNG and H_2) as well as vehicles adapted to specific biofuels (B100, E85).

3. Results

We have used the combined models to analyse several scenarios (Table 1). In the baseline scenario, which is referred to as Implemented Policies (IPO), a possible future for Gauteng has been modelled that not only describes and analyses a business-as-usual development, but also incorporates policies that are currently being discussed and are likely to be implemented in future, such as the Energy Efficiency Strategy (2013), the Integrated Resources Plan 2010 (2013), the National Energy Act 2008 (2008), the White Paper on Renewable Energy (2003) and the Petroleum Products Act (2012) These policies, for example, include targets for biofuel usage as well as installation targets for solar water heating and the renewable energy independent power producer procurement programme.

The principle of GHG mitigation is the main feature of the three further scenarios modelled. These scenarios go beyond the implemented policy targets. Two mitigation scenarios are based on the GIES, for which several plausible mitigation measures have been postulated as well as the mitigation target of reducing GHG emissions by about 50% compared to the reference development in 2055 (DLGH, 2010). One of those scenarios (LGL) considers these objectives to be achieved with the least system cost without favouring any option. The second scenario (LGS) looks at how to achieve the targets with a more ambitious focus to integrate renewable energy options (Table 1).

The third mitigation scenario (LRS) assesses how Gauteng can contribute to the national targets described in the ambitious Long-term Mitigation Scenarios (LTMS), which target the global ambition not to exceed a warming of 2 °C by reducing GHG emissions by about 30–40% in 2050 compared to 2003 levels (Winkler, 2007). These greenhouse gas emission targets are relatively stringent, but are the basis of South Africa's participation in international negotiations on greenhouse gas reductions and the

Table 1
Scenario definition and main scenario parameters.

Scenario name and definition	
IPO	The scenario "Implemented Policies" (IPO) can be described as the reference-scenario. Here, the existing policies are implemented as well as those which are seen to be likely implemented in the future (e.g. a biofuel blending quota). The IPO scenario thus incorporates not only a 'business as usual' case but reflects also the effects of the future policy making process.
LGL	"Least Cost" (LGL), which defines GHG-reduction targets as pointed out in the GIES. The LGL scenario does not incorporate increased targets for using renewable energy but looks at implementing those technologies to achieve the postulated targets with least costs.
LGS	"Solar Province" (LGS) scenario is based on the GHG mitigation targets defined in the GIES as the LGL scenario. In contrast to LGL the LGS scenario aims at a possible development of Gauteng to become a "Solar Province". This includes higher requirements for using renewable energy for electricity and primary energy provision and excluded the use of nuclear energy.
LRS	This scenario refers to the LTMS scenario, which was mandated by the SA Cabinet and was carried out by the Department of Environmental Affairs & Tourism (DEAT). Here the GHG-reduction targets are as required to be in conformity with national mitigation targets and with IPCC reports.
Main scenario parameters	
Energy Carrier Prices	Based on the "current policies scenario" of the World Energy Outlook 2014 (IEA, 2014) (Crude oil price: 132 USD_{2013}/bbl in 2040).
Population	1.8%/a growth; population nearly doubles between 2007 and 2040 (183%)
Households	Average size decreases; number of households more than doubles (215%) between 2007 and 2040
GDP	Annual average growth of 4.1% (2007–2040); share of primary sector of GDP shrinks to a quarter of its size, secondary decreases little and tertiary sector increases to a share of 75%
Employment	Increase of 182%; halving of employment in primary sector, nearly doubling in secondary and more than doubling in tertiary sector
Income distribution	Gini coefficient decreases from 0.57 to 0.50
Electrification	All formal households electrified by 2040; informal households unelectrified share of 40% in 2040
Mobility	Vehicle travel demand increases for road passenger from 115 to 276 bn pkm and for road freight from 61 to 137 bn tkm between 2007 and 2040

basis for an updated informal voluntary pledge under the Copenhagen Accord (2009). As Gauteng is the economic hub of the country, its role in this process needs to be significant.

When analysing these scenarios, the economic framework adopted in this paper was taken to be constant, meaning that the same projection for gross domestic product (GPD), population growth and employment development were used for all scenarios. All assumptions regarding the socio-economic framework have been documented in Wehnert et al. (2010).

Table 2 shows the emission factors and specific fuel consumption figures we created with TEMT for the example of a passenger car. Replacing the base year fleet of 2007 (which virtually only consists of conventional petrol and diesel vehicles) significant reductions in both specific energy consumption and emissions can be achieved. For example, a new passenger car with Euro 5 emission standards is calculated to perform about 9–19% better than the fleet average depending on driving speed and pattern. Specific energy consumption for future vehicle designs might even be reduced further (as well as their specific emissions). In the year 2040, we expect the specific fuel consumption to be reduced by about 20% for diesel passenger cars and about 30% for petrol passenger cars in comparison to modern Euro 5 vehicles. Additional improvements are possible if using alternative powertrains such as hybrid electric vehicles.

Using alternative fuels can also reduce the GHG and pollutant emissions (Table 2). CO_2 emissions for biofuels are balanced to zero based on the assumption that the carbon emitted is bound in the biomass. We assume the same trajectory for reductions in specific fuel consumption and emissions for the future development of combustion engines using alternative fuels (i.e., biodiesel, ethanol, CNG, LPG or hydrogen) as shown for petrol and diesel passenger cars. For battery electric vehicles (BEV) we assume a reduction of specific energy consumption of 20% and for fuel cell hybrid-electric vehicles of 25% between current values and 2040. These figures, presented for passenger cars in Table 2, as well as those for the other transport modes are used as input to the TIMES-GEECO model for the scenario analysis.

The results of the scenario analysis show that in the reference scenario GHG emissions in Gauteng are likely to increase substantially, to about 220 Mt CO_2eq in 2040 (+64%). This is caused by two main factors: the expected increase in population and in GDP, and the demographic shift and increase in personal wealth. As a result, the total final energy consumption increases significantly in most sectors. Under IPO the transport final energy consumption increases significantly from 271 PJ in 2007 to about 503 PJ in 2040 (+86%). This growth is mainly due to aviation, individual passenger and freight transport. As the scenario calculations see the future energy supply still largely based on coal and fossil fuels, significant GHG emissions are produced.

Fig. 3 displays the tank-to-wheel (TTW) GHG emissions caused by the transport activity in Gauteng. Under the conditions of the IPO scenario a significant increase of transport GHG emissions is probable. Whilst in 2007 about 16.1 Mt CO_2eq of TTW GHG emissions were emitted this figure increases by more than 43% to about 23.1 Mt CO_2eq in 2040. It can be identified that this increase is lower than the growth in vehicle activity (+147%), which is foremost as a result of the renewal of the fleet between 2007 and 2040. As old vehicles are replaced with new ones, a significant decrease in specific energy consumption can be identified. Furthermore, the increased utilisation of alternative powertrains additionally supports slowing down the rise in TTW GHG emissions. Thus, average specific GHG emissions of passenger transport reduce from 278 g CO_2eq/km in 2007 to less than 146 g CO_2eq/km in 2040 which equals to a reduction of 48%. In freight transport the corresponding figures are 536 g CO_2eq/km in 2007 and 323 g CO_2eq/km in 2040 (–40%).

The highest reduction of overall TTW GHG emissions in the transport sector is required in the LRS scenario, with about 3.8 Mt CO_2eq (−15%) less compared to the reference in 2040. This reduction is achieved by using a higher proportion of alternative powertrains than the reference scenario, IPO. Furthermore, the fuel provision for the transport sector has to be changed under climate protection targets. Thus, in 2040, the share of biofuels has increased in all mitigation scenarios (up to +39 PJ in LRS). Conversely, the contribution of fossil-based synthetic fuels to the fuel mix is reduced. The highest reduction in synthetic fuel use (−99 PJ) compared to the reference scenario in 2040 is in the LRS scenario, for which there are no synthetic fuels from coal or natural gas in use.

The contribution of alternative powertrains to total vehicle activity is about 10% and 18% higher in the LGL and LGS, respectively, than under IPO. In the LRS scenario in 2040 we can identify

Table 2
Emission factors for passenger cars in Gauteng as result of the TEMT model and input to TIMES-GEECO. Fuel consumption (FC) is shown in MJ/km all others values are given in g/km

Fuel type	Emission concept	Street type	FC	CO_2	CH_4	N_2O	CO	HC	NO_x	PM
Petrol	**Fleet average 2007**	Highway	3.3	234.7	0.034	0.007	9.663	0.987	1.479	0.009
		Rural	3.1	220.9	0.039	0.007	9.829	1.116	1.197	0.006
		Urban	3.7	262.8	0.054	0.009	11.921	1.554	1.122	0.004
	EURO4	Highway	3.0	215.3	0.001	0.002	0.479	0.008	0.058	0.002
		Rural	2.8	202.5	0.000	0.002	0.293	0.006	0.054	0.001
		Urban	3.4	240.9	0.001	0.003	0.264	0.006	0.061	0.001
	EURO5	Highway	2.7	196.2	0.001	0.001	0.393	0.007	0.060	0.002
		Rural	2.6	184.6	0.000	0.001	0.269	0.005	0.053	0.001
		Urban	3.0	212.1	0.001	0.001	0.304	0.006	0.064	0.001
	EURO6	Highway	2.6	182.8	0.000	0.001	0.328	0.006	0.056	0.001
		Rural	2.4	172.0	0.000	0.001	0.260	0.005	0.050	0.001
		Urban	2.7	191.9	0.000	0.001	0.293	0.006	0.061	0.001
	Post-EURO6	Highway	2.3	164.6	0.000	0.001	0.295	0.005	0.050	0.001
		Rural	2.2	154.8	0.000	0.001	0.234	0.005	0.045	0.001
		Urban	2.4	172.7	0.000	0.001	0.264	0.005	0.055	0.001
	2032/2040	Highway	1.9	134.8	0.000	0.001	0.300	0.005	0.036	0.001
		Rural	1.8	126.8	0.000	0.001	0.183	0.004	0.034	0.001
		Urban	2.1	150.8	0.000	0.002	0.165	0.004	0.038	0.000
	Mild Hybrid (EURO5)	Highway	2.4	172.7	0.000	0.001	0.346	0.006	0.053	0.002
		Rural	2.3	162.5	0.000	0.001	0.237	0.005	0.047	0.001
		Urban	2.3	165.4	0.000	0.001	0.237	0.005	0.050	0.001
	Full Hybrid (EURO5)	Highway	2.3	166.8	0.000	0.000	0.334	0.006	0.051	0.002
		Rural	2.2	156.9	0.000	0.001	0.229	0.005	0.045	0.001
		Urban	2.2	159.1	0.000	0.001	0.228	0.005	0.048	0.001
	Plug-in Hybrid (EURO5)	Highway	1.6	33.4	0.000	0.000	0.067	0.001	0.010	0.000
		Rural	1.6	31.4	0.000	0.000	0.046	0.001	0.009	0.000
		Urban	1.0	31.8	0.000	0.000	0.046	0.001	0.010	0.000
Diesel	**Fleet average 2007**	Highway	2.3	166.5	0.002	0.003	0.431	0.095	0.769	0.104
		Rural	2.3	170.3	0.003	0.003	0.467	0.122	0.708	0.111
		Urban	2.7	199.1	0.004	0.004	0.662	0.185	0.747	0.118
	EURO4	Highway	2.1	153.4	0.000	0.005	0.036	0.010	0.551	0.001
		Rural	2.1	156.9	0.000	0.005	0.038	0.012	0.542	0.002
		Urban	2.5	184.1	0.000	0.007	0.064	0.015	0.625	0.002
	EURO5	Highway	2.0	149.0	0.000	0.005	0.025	0.010	0.545	0.001
		Rural	2.1	152.3	0.000	0.005	0.027	0.012	0.538	0.002
		Urban	2.3	170.2	0.000	0.007	0.045	0.016	0.622	0.002
	EURO6	Highway	1.9	141.1	0.000	0.005	0.036	0.009	0.189	0.001
		Rural	2.0	144.3	0.000	0.005	0.037	0.011	0.189	0.002
		Urban	2.2	159.5	0.000	0.007	0.063	0.014	0.219	0.002
	Post-EURO6	Highway	1.7	127.0	0.000	0.005	0.032	0.008	0.170	0.001
		Rural	1.8	129.9	0.000	0.005	0.034	0.010	0.170	0.001
		Urban	2.0	143.5	0.000	0.006	0.057	0.013	0.197	0.002
	2032/2040	Highway	1.6	116.6	0.000	0.004	0.028	0.008	0.419	0.001
		Rural	1.6	119.2	0.000	0.004	0.029	0.009	0.412	0.001
		Urban	1.9	139.9	0.000	0.005	0.048	0.012	0.475	0.002
	Mild Hybrid (EURO5)	Highway	1.8	135.6	0.000	0.005	0.023	0.009	0.496	0.001
		Rural	1.9	138.6	0.000	0.005	0.024	0.011	0.490	0.001
		Urban	1.9	137.9	0.000	0.006	0.036	0.013	0.504	0.002
	Full Hybrid (EURO5)	Highway	1.8	131.1	0.000	0.005	0.022	0.009	0.480	0.001
		Rural	1.8	134.0	0.000	0.005	0.023	0.011	0.474	0.001
		Urban	1.8	132.8	0.000	0.005	0.035	0.012	0.485	0.002
	Plug-in Hybrid (EURO5)	Highway	1.5	26.2	0.000	0.001	0.004	0.002	0.096	0.000
		Rural	1.5	26.8	0.000	0.001	0.005	0.002	0.095	0.000
		Urban	0.9	26.6	0.000	0.001	0.007	0.002	0.097	0.000
Electricity	**BEV**	Highway	1.4							
		Rural	1.4							
		Urban	0.7							
Biodiesel	**B100 (EURO4)**	Highway	2.1	0.0	0.000	0.005	0.018	0.004	0.606	0.001
		Rural	2.1	0.0	0.000	0.005	0.019	0.004	0.596	0.001
		Urban	2.5	0.0	0.000	0.007	0.032	0.005	0.687	0.001
Ethanol	**E85 (EURO4)**	Highway	2.8	42.8	0.001	0.001	0.384	0.007	0.047	0.001
		Rural	2.7	41.3	0.001	0.001	0.234	0.005	0.043	0.001
		Urban	3.2	48.9	0.001	0.002	0.211	0.005	0.049	0.000
Hydrogen	**H2ICEV (EURO4)**	Highway	2.9	0.0	0.000	0.001	0.005	0.004	0.451	0.000
		Rural	2.9	0.0	0.000	0.001	0.005	0.004	0.424	0.000
		Urban	2.6	0.0	0.000	0.001	0.006	0.004	0.505	0.000

Table 2 (*continued*)

Fuel type	Emission concept	Street type	FC	CO_2	CH_4	N_2O	CO	HC	NO_x	PM
	FCHEV	Highway	2.1							
		Rural	2.1							
		Urban	1.5							
CNG	**EURO4**	Highway	3.0	177.2	0.013	0.001	0.134	0.157	0.067	0.004
		Rural	2.8	166.7	0.010	0.001	0.082	0.119	0.063	0.002
		Urban	3.4	198.3	0.011	0.001	0.074	0.126	0.071	0.001
LPG	**EURO4**	Highway	2.8	169.5	0.001	0.002	0.288	0.008	0.052	0.001
		Rural	2.6	159.4	0.000	0.002	0.176	0.006	0.048	0.000
		Urban	3.1	189.7	0.001	0.003	0.159	0.006	0.055	0.000

a total activity of non-conventional vehicles of about $36.8 \cdot 10^9$ vkm which is more than twice the figure of the IPO scenario for the same year where about $15.3 \cdot 10^9$ vkm were performed with alternative powertrains. These include mainly E85-fuelled vehicles and mild hybrid vehicles which contribute about $19.1 \cdot 10^9$ vkm (18%) and $14.6 \cdot 10^9$ vkm (14%) to total vehicle activity in the LRS scenario.

The increased contribution of alternative powertrains as well as the substitution of fossil fuels results in increasing mobility costs on a moderate level. In the LGL and LGS scenarios the mobility costs (measured in ZAR/km) are in average about 2–7% (i.e. 0.04–0.16 ZAR/km) higher than in the IPO scenario. In the LRS scenario, which reflects the most stringent GHG mitigation targets, the increase in mobility costs is up to 13% (i.e. 0.32 ZAR/km) in the year 2040.

The large variety of output data regarding vehicle travel and fuel consumption generated from the TIMES-GEECO model for the different scenarios was displayed spatially through the TEMT model. Johannesburg, including the city centre, the inner ring road and the inner suburbs, was chosen to visualize vehicle emissions as it is an area with high-traffic-density in Gauteng. The emissions are presented for a raster of a 4 km^2 grid on a typical working day for a sample part of the city (Fig. 4). While in TIMES-GEECO the focus is on GHG reduction measures, in TEMT the analysis emphasises air pollutants, such as NO_x emissions.

The spatial view of the results of the calculation shows where the transport energy consumption and emissions take place. The hot-spots with the highest transport emission levels were identified in the Johannesburg Central Business District (CBD), in Selby, and along the southern and eastern bypasses, with average values of more than 0.25 t NO_x/day/km^2. These hot-spots are caused by vehicle overload of the transport network arising from urban vehicle driving conditions, such as stop-and-go driving and congestion at peak hours. These driving conditions increase the fuel consumption of each vehicle.

Comparing emissions in future with those in the base year show a significant reduction for all pollutants in all scenarios (Fig. 4). This is due to the replacement of the province's ageing baseline vehicle fleet with an increasing share of vehicles subject to higher emission standards. Between the IPO and mitigation scenarios we have not calculated a significant difference in total pollutant emissions as the difference between the modern vehicle technologies are outweighed by the improvement in comparison the base year's fleet. Hydrocarbon and CO emissions decline in average by about 94–98% in 2040 compared to the base year of 2007. NO_x emissions fall from 101 kt in 2007 to 42 kt in IPO and to 40 kt in the three mitigation scenarios. The reduction of SO_2 emissions between 2007 and 2040 is between 77% and 85%. Most of these reductions are likely until the year 2030 when the baseline fleet of 2007 is replaced with new vehicles. After 2030, non-GHG emissions increase slightly again in the high socio-economic growth scenario, due to the increase in vehicle travel. The main reductions of pollutant emissions can be found in the city centre and especially along the high-traffic-density highways and urban roads.

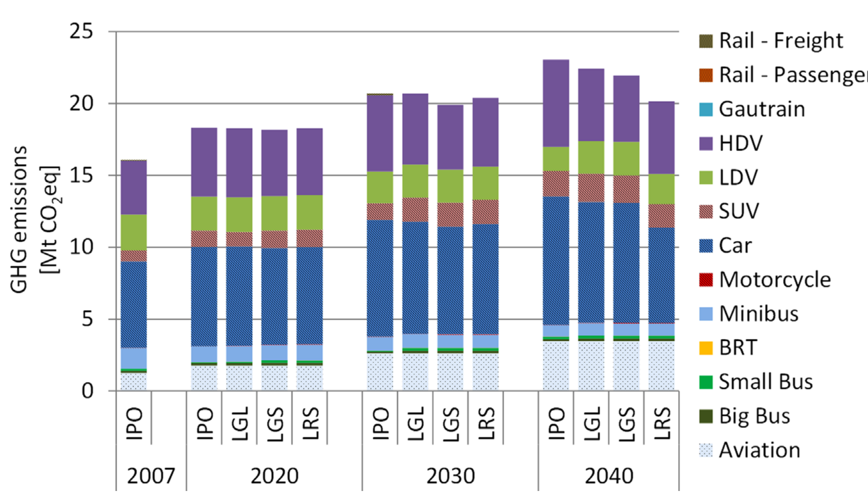

Fig. 3. Transport GHG emissions under GHG mitigation targets in comparison to the IPO scenario.

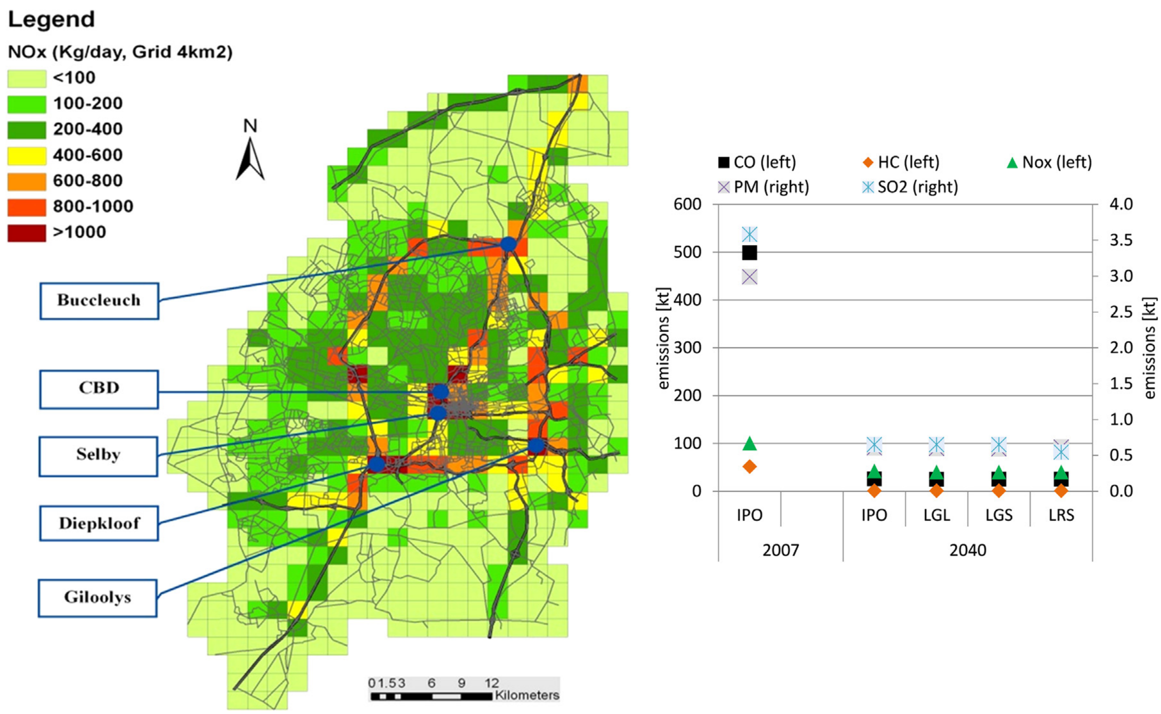

Fig. 4. Illustration of the application of the TEMT model to the city of Johannesburg nitrogen oxides road transport emissions in Johannesburg in the year 2009 (left) and pollutant emissions between 2007 and 2040 in the IPO scenario (right).

4. Conclusion and policy implications

4.1. Conclusion

We have developed two transport oriented models in order to demonstrate how the transport sector can be assessed within an energy system taking into consideration locational information (GIS). The models have been applied to Gauteng Province in South Africa to derive necessary data on vehicle performance and prospects of alternative powertrain technologies as well as to analyse the future development of the transport sector for different scenarios. Modelling results have been interpreted to give policy recommendations towards a more sustainable development for the province.

The TEMT model generates the emission factors and specific energy consumption of vehicles by calibrating the information on the characteristics of vehicle fleet in terms of the Handbook of Emission Factors for Road Transport (HBEFA). This information is then incorporated into the TIMES-GEECO energy system model. Comparing different scenarios using TIMES-GEECO leads to robust options (solutions appearing in most of the scenarios independently from uncertain future development) which can be seen as suitable measures for reaching (for example, climate) targets under a set of various theoretical futures. The information generated was then fed into the GIS software to display the results spatially.

Scenario calculations showed that pollutant emissions in Gauteng can be significantly reduced by renewing the present vehicle fleet in Gauteng that consists of comparably old vehicles with newer ones compliant to current European emission standards. Whereas we see pollutant emissions to be reduced in future in Gauteng, the increase in traffic causes a significant uptake in energy demand and a high level of greenhouse gas emissions even under climate protection targets. However, the analysis shows that switching to alternative fuels and vehicle technologies can reduce the GHG emissions in the transport sector significantly. In an urban environment such as Gauteng, hybrid vehicles can substantially improve energy efficiency most effectively if applied to vehicles with a high urban mileage such as buses.

The presented modelling approach required a lot of detailed data. The most uncertain aspect of modelling remains the assignment of emission standards to particular vehicles and whether they are equipped with a catalytic converter. Furthermore, the TEMT model does not cover passenger travel demand originating from educational and leisure activities in the province, since statistical data is only available on working trips, which are the biggest part of travel demand. It can be also noted that the results of modelling, i.e. estimation of emissions from road transport, depends largely from the correct estimation of the vehicle volumes. The MATSim modelling was performed in order to adjust the travel demand to the most recent statistics available. Future measurements can help to better calibrate the model to real conditions. Future research should compare the calculated figures with real-world measurements.

The distribution of the emissions from road transport on the streets which are not covered by the current MATSim road network (based on OpenStreet Map data) remains still to be investigated. Specifically the outskirts of the link network of Gauteng Province are not depicted in details. However, it is expected that the amount of vehicle volumes in those areas, and thus transport emissions calculated, will not have a significant impact on the emission inventory modelling results, since the vehicle volumes are expected to be small and are thus omitted in this study.

4.2. Policy implications

4.2.1. Alternative fuels

Biofuels, such as biodiesel from waste cooking oil and, in the future, ethanol from lignocellulosic biomass, can reduce GHG emissions cost effectively. For the provision of biodiesel from waste cooking oil, the oil supply has to be ensured as available resources are currently used for animal food. At first, the fuel can be used for governmental vehicles to demonstrate feasibility. A bylaw or blending quota can be the legal basis for a large scale roll-out. Furthermore, an awareness and communication campaign will ensure the communal and public support for the new fuel.

4.2.2. Stricter emissions legislation

Stricter emission legislation should be enforced in order to significantly reduce pollutant emission. A fleet renewal could support and accelerate this process as large numbers of comparably old vehicles are currently being used in Gauteng.

4.2.3. Alternative powertrains

To support the usage of alternative powertrains a showcase for public buses could be created, especially where investments in new vehicles are undertaken anyway, e.g., for the bus rapid transit (BRT) systems. An incentive programme can be the basis for implementation of hybrid vehicles for minibuses. A financing mechanism could be initiated which covers the higher investment costs and splits the savings due to reduced fuel consumption between operators and financers.

4.2.4. Improved data statistics and monitoring

It can be recommended to the South African stakeholders and policymakers to restructure the NATIS system for vehicle registration and to include the information on the emission standard and technology of the vehicles. This will help to improve the estimates of the transport emissions in Gauteng Province and South Africa in the future more precisely and to elaborate effective mitigation actions for GHG emission reductions more effectively. In addition, it can be recommended to establish an online monitoring system in order to track on a regular basis vehicle flows of each vehicle category in order to improve the knowledge on the transport demand in the province.

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